

## Review article

## The impact of pollen-related food allergens on pollen allergy

Patients with birch pollen allergy frequently develop hypersensitivity reactions to certain foods, e.g. apples, celery, carrots and hazelnuts. These reactions are mainly caused by IgE-antibodies specific for the major birch pollen allergen, Bet v 1, which cross-react with homologous proteins in these foods. Analyzing the T-cell response to Bet v 1-related food allergens revealed that these dietary proteins contain several distinct T-cell epitopes and activate Bet v 1-specific T cells to proliferate and produce cytokines. Several of these cross-reactive T-cell epitopes were not destroyed by simulated gastrointestinal digestion of food allergens and stimulated Bet v 1-specific T cells despite nonreactivity with IgE antibodies. Similarly, cooked food allergens did not elicit IgE-mediated symptoms (oral allergy syndromes) but caused T-cell-mediated late-phase reactions (deterioration of atopic eczema) in birch pollen-allergic patients with atopic dermatitis because thermal processing affected their conformational structure and not the primary amino acid sequence. Thus, T-cell cross-reactivity between Bet v 1 and related food allergens occurs independently of IgE-cross-reactivity *in vitro* and *in vivo*. We speculate that symptom-free consumption of pollen-related food allergens may have implications for the pollen-specific immune response of allergic individuals.

**B. Bohle**

Department of Pathophysiology, Center for Physiology and Pathophysiology, Medical University of Vienna, Vienna, Austria

Key words: Bet v 1; birch pollen allergy; cross-reactivity; food allergy; T cells.

Barbara Bohle PhD

Department of Pathophysiology  
Center for Physiology and Pathophysiology  
Medical University of Vienna  
Waehringer Guertel 18-20  
AKH-3Q  
A-1090 Wien  
Austria

Accepted for publication 20 September 2006

Food allergy represents an important manifestation of atopic allergy. Primary food allergy mainly affects young children whereas adults frequently develop food allergy as a consequence of an inhalant sensitization. Pollen-related food allergy has become the most frequent form of food allergy in adolescent and adult individuals in Europe. A typical example is the 'birch-fruit-vegetable-syndrome'. More than 70% of birch pollen-allergic individuals develop allergic reactions to stone-fruits, nuts or certain vegetables. In the majority of patients these reactions are confined to the oropharynx and summarized as 'oral allergy syndrome' (OAS) (1). Nevertheless, systemic IgE-mediated reactions such as urticaria, asthma or anaphylactic shock occur occasionally (2, 3). The major birch pollen allergen, Bet v 1, is the most relevant sensitizing protein causing this type of food allergy (4–8) but minor allergens such as Bet v 2, Bet v 5 and Bet v 6 have also been shown to be involved (6–10). Whereas these allergens are recognized by 10–32% of birch pollen-allergic patients, more than 95% display IgE against Bet v 1. The major birch pollen allergen belongs to the pathogenesis-related (PR) protein family 10 (11). Other members of this protein family are present in various foods, such as fruits of *Rosaceae* (e.g. Mal d 1 in apple, Pru a 1 in cherry, Pyr c 1 in pear), vegetables of *Apiaceae* (e.g. Api g 1 in celery, Dau c 1 in carrot), hazelnut (Cor a 1), soybean (Gly m 4), mungbean (Vig r 1) and peanut (Ara h 8) (9, 12–21). These proteins share a high degree of amino acid sequence similarity with the major birch pollen allergen

resulting in a similar tertiary structure (22). Thus, Bet v 1-specific IgE antibodies can bind to these dietary proteins which may cause immediate hypersensitivity reactions upon consumption of the respective foods. The IgE-cross-reactivity between Bet v 1 and food homologues correlates roughly with the similarity of their primary protein structures. Hence, Bet v 1-specific IgE antibodies react preferentially with allergens in *Rosaceae* fruits sharing between 56% and 59% of amino acid similarity with Bet v 1 and less frequently with homologues in vegetables of the *Apiaceae* family sharing 37–41% (23, 24). Analyzing the IgE binding of 50 Bet v 1-positive patients to different food allergens we observed that 99% reacted with Mal d 1, 93% with Cor a 1, 59% with Api g 1 and 38% with Dau c 1. Vice versa, patients with birch pollen-related food allergy are predominantly sensitized to Bet v 1-homologues and less frequently recognize other allergens contained in both sources, e.g. profilins (14, 25, 26).

In accordance with the continuously increasing prevalence of birch pollinosis during the past 15 years, birch pollen-related food allergy has become an important disorder perennially impairing the quality of life of allergic patients including the time period outside the pollen season (27). However, at present no efficient therapy for this type of food allergy is available. One reason for this lack is obviously the still limited knowledge about mechanisms operative in this allergic disorder. By adjusting the experimental methodology which has previously been successfully applied for allergens from sources other than

food, we characterized the T-cell response underlying pollen-associated food allergy (28–31). Our experimental approach is based on the isolation and expansion of allergen-specific T-cell lines (TCL) and clones (TCC) from the peripheral blood of humans which then serve as perfect tools to analyze the phenotype of specific T cells, their cytokine response to specific stimulation as well as their epitope recognition and MHC restriction (Fig. 1). Up to now, T-cell responses to three relevant Bet v 1 homologues from different botanical families, namely Mal d 1 from apple, Api g 1 from celery and Cor a 1 from hazelnuts were analyzed in detail (32–34). The majority of food-specific TCC belonged to the Th2 subset, which produced high amounts of IL-4 but only little or no IFN- $\gamma$  upon allergen-specific stimulation. By stimulating food-specific TCL and TCC with a panel of synthetic 12-mer peptides

overlapping for nine residues and representing the entire amino acid sequence of each allergen, T-cell epitopes in Mal d 1, Api g 1 and Cor a 1 were identified. Similar to the major birch pollen allergen, its dietary homologues contained multiple T-cell-reactive determinants (35). These T-cell epitopes were located in parts corresponding to relevant T-cell-activating regions of Bet v 1 (Fig. 2). Accordingly, the majority of food-reactive T-cell cultures also responded to stimulation with Bet v 1 demonstrating that these T-cell epitopes were indeed cross-reacting. Interestingly, most T-cell cultures induced with Mal d 1- and Api g 1 proliferated by far more pronounced in response to Bet v 1 than to the food proteins. These cultures also lost their reactivity with the food allergens after a few expansion cycles *in vitro* but remained highly Bet v 1-reactive. These results strongly indicated that

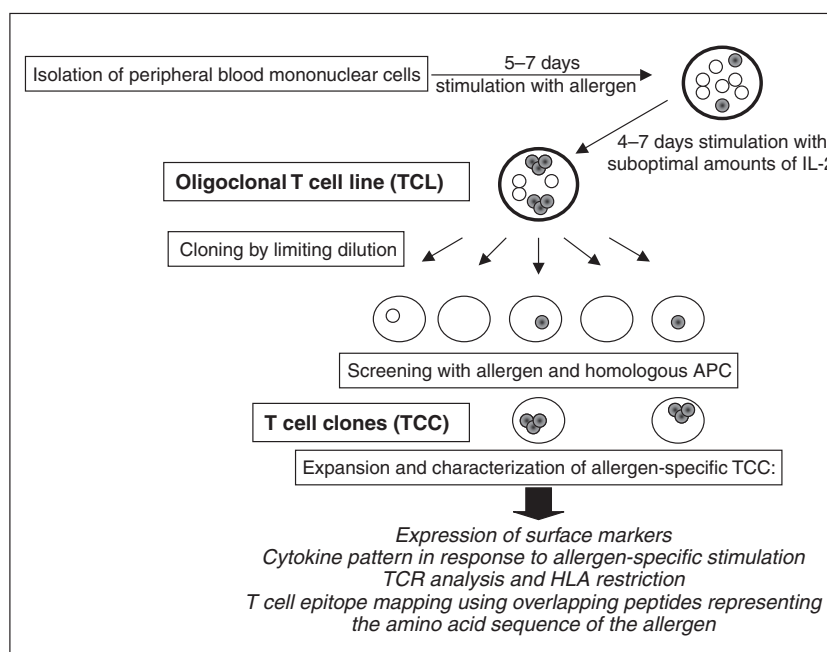


Figure 1. Experimental protocol to isolate allergen-specific T lymphocytes. For explanation see text.

|         |     |        |                  |                  |          |                |             |            |       |     |
|---------|-----|--------|------------------|------------------|----------|----------------|-------------|------------|-------|-----|
| Bet v 1 | 1   | GVFN   | YETETTSVI        | FAARLFKAFILDG    | DNLF     | PKVAPQA        | ISSVENIE    | GNGGPGTIK  | KISFP | 60  |
| Mal d 1 | 1   | GV+    | +E E TS IP       | +RLFKAF+LD DNL   | PK+APQAI | E              | +EGNGGPGTIK | KI+F       |       | 60  |
| Api g 1 | 1   | GV     | + E TS +         | A ++F+ F++D D +  | PK AP A  | SVE I          | +G+GGPGT+K  | I+ P       |       | 59  |
| Bet v 1 | 61  | EGFP   | FKYVKDRVDEVD     | HTNFKYNYSVIEGGPI | GD       | TLKISNEIKIVATP | D           | GGSILKISN  |       | 119 |
| Mal d 1 | 61  | EG     | + YVK R+D +D     | ++ Y+Y++IEG +    |          | DT+EKIS E K+VA |             | G +I IS+   |       | 119 |
| Api g 1 | 60  | +G P   | + R+D V+         | ++YSVI+G +       |          | +E I N + +V T  |             | DGGS I K + |       | 119 |
| Bet v 1 | 120 | KYHTK  | GDHEVKAEQVKASKEM | GETLLRAVESYLLAHS | DAYN     |                |             |            |       | 160 |
| Mal d 1 | 120 | YHTKG+ | E+K E VK KE      | L + +ESYL H      | DAYN     |                |             |            |       | 160 |
| Api g 1 | 120 | +HTKGD | V E +K + E       | L +A+E+YL+A+     |          |                |             |            |       | 154 |

Figure 2. Relevant T-cell epitopes of the major birch pollen allergen involved in cross-reactivity with food allergens. Bet v 1-epitopes shared with the major apple allergen are shown in dark grey boxes, epitopes shared with the major celery allergen in white boxes and epitopes shared with both food allergens in light grey boxes. The location of the dominant T-cell epitopes of each food allergen is underlined in the respective amino acid sequence.

T cells specific for these food allergens were in fact pollen-specific, cross-reactive clonotypes. Thus, these findings at the T-cell level corroborate the concept that sensitization to PR-10 like food allergens develops as consequence of primary inhalant sensitization to the major birch pollen allergen.

### Food allergens are potent activators of pollen-specific T cells

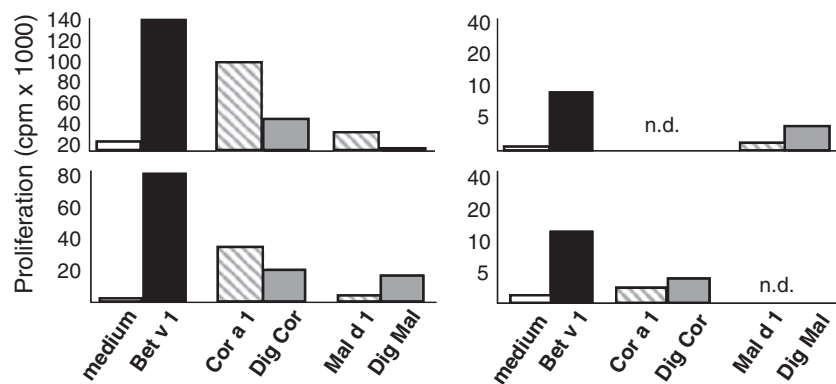
To evaluate whether pollen-related food allergens were able to activate pollen-specific T cells, Bet v 1-specific TCL and TCC recognizing different T-cell epitopes were stimulated with Mal d 1 (apple), Api g 1 (celery) or Cor a 1 (hazelnut) (32–34). Most cultures responded with proliferation and cytokine secretion showing that Bet v 1-related food allergens activated Bet v 1-specific T cells with diverse specificities. To elucidate whether particular epitopes of Bet v 1 were predominantly involved in T-cell reactivity with each food homologue, more than 50 Bet v 1-specific TCL with known epitope specificity from different birch pollen-allergic patients were screened for reactivity with Mal d 1 or Api g 1 (36). Epitope recognition patterns of food allergen-reactive and nonreactive TCL were compared and the percentage of peptide-reactive cultures was calculated for each group. Cross-reactivity between Bet v 1 and Mal d 1 was predominantly restricted to one single T-cell epitope, Bet v 1<sub>142–153</sub>. This finding was surprising because both allergens share 71% of amino acid sequence similarity (12). On the other hand, T-cell cross-reactivity between Bet v 1 and Api g 1 was associated with several different epitopes spreading the entire amino acid sequence of the pollen allergen although both allergens share only 61% of sequence similarity (15). It has to be emphasized that all individuals (100%) studied had an OAS to apple whereas only around 35% of these patients experienced immediate IgE-mediated reactions to celery. Hence, the degree of amino acid sequence similarity between Bet v 1 and its dietary homologues correlates roughly with the occurrence of food allergy but not with the magnitude of cross-reactivity at the T-cell level. These findings support the necessity for the characterization of the T-cell response including the identification of T-cell epitopes in allergens to understand immunological mechanisms underlying different manifestations of atopic diseases.

Analyzing the epitope recognition pattern in Bet v 1-specific TCL generated from a large number ( $n = 57$ ) of different birch pollen-allergic patients, one immunodominant T-cell epitope of Bet v 1, recognized by 61% of the individuals, was identified (35). This peptide which comprises amino acid residues 142–156 is located in the C-terminal part of the major birch pollen allergen, a region highly conserved among different PR-10 like proteins. When TCL and TCC specific for this immunodominant epitope were stimulated with PR-10 like

proteins from apple, cherry, pear, peach, hazelnut, celery, carrot or soybean, the majority of cultures responded to more than one food allergen by proliferation and production of IL-4 and IL-5 (35). Hence, Th2 cells specific for the most frequently recognized T-cell epitope of Bet v 1 can be easily activated by various birch pollen-related foods to proliferate and synthesize cytokines.

### Gastrointestinal digestion does not destroy the T-cell-activating capacity of pollen-related food allergens

It has been previously shown that PR-10-like proteins undergo rapid gastric degradation leading to the immediate loss of their IgE-binding capacity (37, 38). In contrast to the IgE-binding sites of these proteins, which are conformational epitopes depending on the tertiary protein structure (22, 39), T-cell epitopes are short, linear peptides (32–34). However, such peptides may survive gastrointestinal degradation and remain stimulatory for T lymphocytes (40). To evaluate this hypothesis, Mal d 1 (apple), Api g 1 (celery) and Cor a 1 (hazelnut) were incubated with pepsin followed by trypsin to simulate gastrointestinal degradation. Thereafter, digested allergens were tested for IgE binding, their capacity to induce mediator release and T-cell activation in peripheral blood mononuclear cells (PBMC) as well as in Bet v 1-specific TCL and TCC. Although allergens had completely lost their capacity to bind IgE and to induce mediator release after incubation with enzymes their products still induced proliferation in PBMC from allergic and nonallergic individuals. Enzymatically digested Mal d 1 and Cor a 1 also activated Bet v 1-specific TCL and TCC to proliferate and to synthesize similar cytokines comparable with stimulation with entire food allergens or Bet v 1 (Fig. 3). Different allergen fragments created by enzymatic degradation were identified and found to match relevant T-cell epitopes of Bet v 1. TCC specific for these Bet v 1 epitopes responded to digested Mal d 1 and Cor a 1, confirming that their proteolytic fragments indeed contained epitopes cross-reactive with pollen-specific T cells. Taken together, exposure of PR-10 like food proteins to gastrointestinal proteases does completely abolish their ability to bind IgE but not their capability to stimulate Bet v 1-specific T cells. These findings provide a biochemical and immunological mechanism underlying the clinical observation that ingestion of birch pollen-related foods can trigger the worsening of atopic eczema in patients with cutaneous T-cell responses to Bet v 1 (41). After passing stomach and duodenum, fragments of pollen-related dietary allergens remain that can activate Bet v 1-specific T cells to upregulate homing receptors, migrate to the skin and induce effector reactions. These T-cell-mediated late reactions in the skin seem to be elicited even independently of an IgE-mediated mechanism, similar to late phase reactions which have been shown in the lung of cat-asthmatic patients after subcutaneous administration of peptides



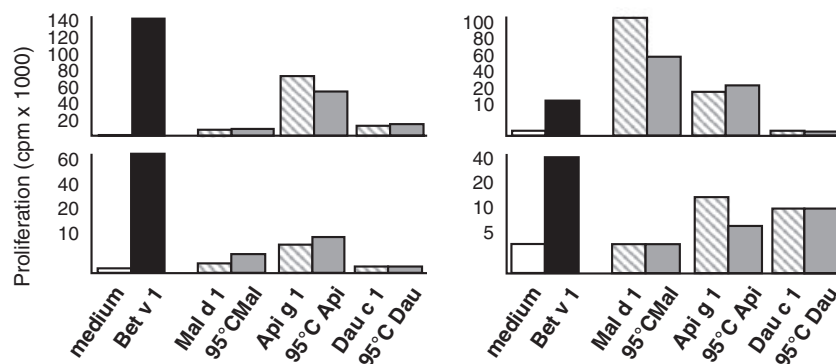
**Figure 3.** Bet v 1-specific T cells respond to food allergens after simulated gastrointestinal digestion. Cor a 1 (hazelnut) and Mal d 1 (apple) were incubated for 30 min with pepsin followed by 30 min of incubation with trypsin. Thereafter, Bet v 1-specific T-cell cultures were stimulated with entire and digested allergens in the presence of autologous peripheral blood mononuclear cells. Bet v 1 and medium alone served as positive and negative control, respectively. After 48 h proliferation (counts per minutes) was determined.

from the major cat allergen, Fel d 1 (42). In accordance with the Fel d 1-peptides, none of the fragments of gastrointestinal-digested Bet v 1-related food allergens was able to bind IgE or to induce basophil degranulation (40).

### T-cell cross-reactivity occurs independently of IgE cross-reactivity

PR-10 like food allergens are considered to be easily destroyed by thermal processing as birch pollen-allergic individuals can generally consume cooked birch pollen-related foods without difficulty. Therefore, allergologists and guide books recommend heating of these aliments to make them safe for consumption (5, 21, 43–45). However, thermal processing of proteins generally causes conformational changes – which in the case of an allergen may result in diminished IgE-binding capacity – but does not affect the primary protein structure and thus, linear T-cell epitopes. Therefore, we speculated that cooked birch pollen-related food will not induce immediate reactions

but may still cause T-cell-mediated symptoms. To prove this hypothesis, Bet v 1-homologues from apple (Mal d 1), celery (Api 1) and carrot (Dau c 1) were exposed to increasing temperatures. Thermal processing changed the conformational protein structure resulting in abrogated IgE-binding and mediator-releasing capacity after incubation at  $>80^{\circ}\text{C}$  for a few minutes (36). However, allergens cooked for 60 min at  $95^{\circ}\text{C}$  still activated PBMC and Bet v 1-specific T-cell cultures to proliferate and produce Th2 cytokines comparably to nonheated proteins (Fig. 4). To substantiate these results *in vivo* and with natural allergens, Bet v 1-mono-sensitized patients with OASs and atopic dermatitis were tested in double-blind placebo-controlled food challenges (DBPCFC) with fresh and cooked apple, celery and carrot, respectively (36). The fact that cooked foods did not induce an OAS in any of the patients confirmed that heated Bet v 1-related allergens in their natural food matrix lose the ability to induce IgE-mediated immediate reactions *in vivo*. On the other hand, ingestion of cooked foods triggered a comparable worsening of atopic eczema as fresh foods.



**Figure 4.** Bet v 1-specific T cells respond to cooked food allergens. Mal d 1 (apple), Api g 1 (celery) and Dau c 1 (carrot) were incubated for 60 min at  $95^{\circ}\text{C}$ . Thereafter, Bet v 1-specific T cell cultures were stimulated with native and heated allergens in the presence of autologous PBMC. Bet v 1 and medium alone served as positive and negative control, respectively. After 48 h proliferation (counts per minute) was determined.

As already mentioned, late eczematous reactions in birch pollen-allergic patients observed after consumption of fresh pollen-related foods have been associated with an increased expression of cutaneous lymphocyte antigen of peripheral blood T cells and the presence of Bet v 1-reactive T cells in the isolated skin lesions (41). Our recent *in vitro* and *in vivo* results extend these previous findings by showing that (i) cooked birch pollen-related foods induce similar T-cell activation and T-cell-mediated symptoms as fresh aliments and (ii) T-cell-mediated reactions occur also in the absence of IgE-binding and IgE-mediated reactions. Therefore, pollen-related foods must be considered as important stimuli for pollen-specific T cells even in patients without immediate clinical symptoms of food allergy.

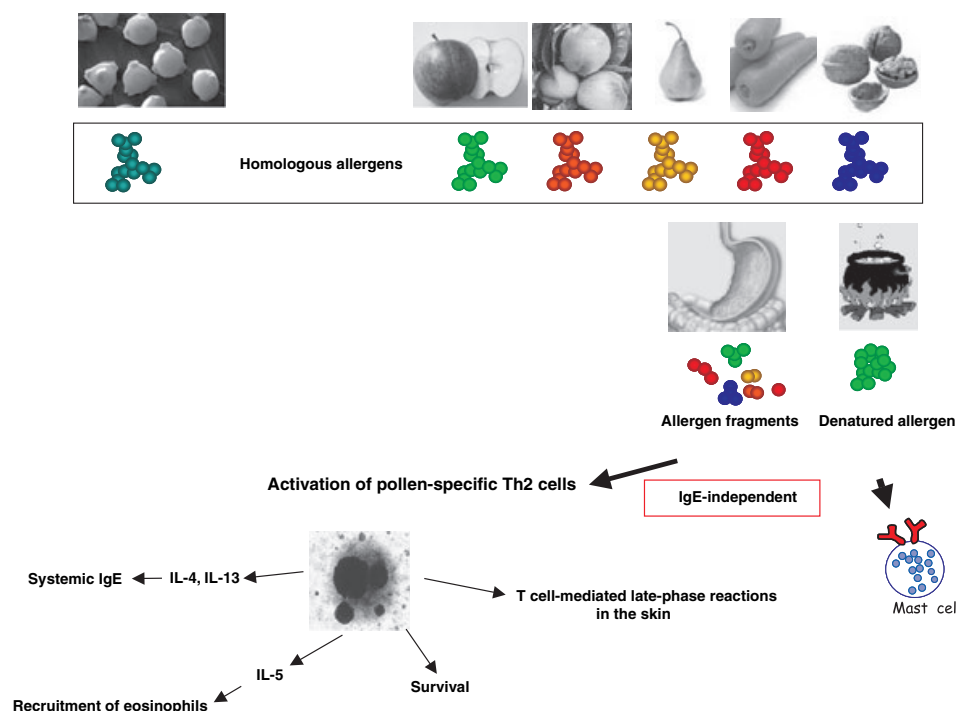
### New treatment strategies for food allergy

Food allergy in adolescents and adults seems to develop as a consequence of primary sensitization to aero-allergens and subsequent immunological cross-reactivity with homologous proteins in foods. This hypothesis is supported by our findings that atopic adults who severely reacted after consumption of millet-containing dishes had been previously exposed to millet dust (46). The majority of birch pollen-allergic patients also experience allergic reactions to foods after having developed pollinosis and Bet v 1 contains most relevant B and T-cell epitopes of its homologous food allergens which is not the case vice versa. Provided that Bet v 1 is the sensitizing allergen for food allergy, one would expect that successful specific immunotherapy (SIT) of pollen allergy should also abolish allergic reactions to foods. As this issue is still controversial (47–51), we conducted a clinical trial analyzing the clinical and immunological effects of sublingual immunotherapy (SLIT) with birch pollen on concomitant allergy to apple. We speculated that the administration of pollen extract directly at the site of food-induced allergic symptoms, i.e. the OAS, may increase the therapeutic efficacy. Furthermore, SLIT is a safe and convenient alternative to subcutaneous specific immunotherapy that has been demonstrated to improve significantly birch pollen-induced rhinitis and asthma (52–54). After 1 year of SLIT, patients showed improvement in nasal challenges with birch pollen. However, similar to reports on subcutaneous treatment with birch pollen, not all patients improved allergic reactions to apples as determined in DBPCFC (47, 49–51). In addition, we pioneered in monitoring the T-cell response to Bet v 1 and Mal d 1 in each individual revealing that only the pollen-specific T-cell response was significantly down regulated by SLIT. From these data we deduce that food-specific clonotypes exist, which escape the induction of peripheral tolerance during pollen-SIT. In line with this hypothesis, we have previously observed that a minority of Mal d 1- or Api g 1-specific TCC did not cross-react with Bet v 1 (32, 33).

Furthermore, analyzing the T-cell response to the Bet v 1-homologue in hazelnuts revealed that several TCC specific for the most frequently recognized T-cell epitope of Cor a 1 did respond to stimulation neither with the major birch pollen nor the hazel pollen allergen (34). Down-regulation of the allergen-specific T-cell response has been shown to be an important component of successful SIT (55). The existence of exclusively food-reactive T cells that are not modulated by vaccination with pollen allergens may provide an immunologic explanation for the noneffectiveness of pollen-therapy on associated food allergy. Therefore, our current and future projects include studies to develop approaches for food allergy treatment, which target all food-specific T lymphocytes. As food extracts are difficult to produce and standardize, we will evaluate the use of recombinant food allergens and/or 'hypoallergenic' variants thereof. Recently, a mutated Mal d 1 molecule lacking most important IgE-binding sites but still containing relevant T-cell epitopes was created (56).

### Conclusion

Only studying the T-cell response to typical aero-allergen-associated food allergens revealed that pollen-related foods have an important impact on pollen-allergy (illustrated in Fig. 5). After consumption, pollen-related food allergens stimulate pollen-specific T cells to exert effector functions. This capability is neither abolished by gastrointestinal digestion nor by cooking of the respective foods, two treatments which abolish immediate IgE-mediated reactions. Thus, T-cell cross-reactivity between pollen and related food allergens is strikingly different at the B- and T-cell level and relevant T-cell-mediated clinical reactions can occur independently from IgE-mediated symptoms. Accordingly, patients who do not suffer from food allergy consume the respective aliments and provide antigens for pollen-specific T lymphocytes. This ingestion may cause 'visible' clinical symptoms, e.g. the deterioration of atopic eczema, but may also solely result in a 'hidden' boost of pollen-specific immune responses. It is known that inhalation of pollen allergens during the pollen season induces local and systemic IgE synthesis and CD4<sup>+</sup> memory T cells (57–59). However, outside the pollen season specific serum IgE levels are typically maintained at increased levels although no allergens are encountered. The consumption of birch pollen-related foods activates pollen-specific Th2 cells *in vivo* also outside the pollen season, thus providing a continuous activation necessary for long-term maintenance of pollen-specific Th2 cells (60, 61). Moreover, perennially activated pollen-specific Th2 cells continuously synthesize high amounts of IL-4, which together with small concentrations of non-degraded, IgE-binding pollen-related food allergens absorbed via the oral and/or gastrointestinal mucosa, could then be sufficient to provoke IgE synthesis in B cells outside the pollen season (62, 63). We conclude that ingestion of pollen-related food



**Figure 5.** Impact of pollen-related food on pollen allergy. Pollen and particular foods contain homologous allergens. Pollen-specific T cells become activated by pollen allergens inhaled during the pollen season but also by native and heat-denatured food homologues or their fragments after gastrointestinal digestion whenever they are consumed. The activation can occur irrespective of IgE-mediated reactions to the food and induce different effector functions, migration to target organs and longevity.

allergens could maintain perennially increased allergen-specific IgE levels in patients suffering from seasonal allergic disorders. These new aspects regarding the induction and maintenance of pollen and associated food allergy are important for the development of efficient vaccines for prevention and therapy of these allergic disorders. Finally, the view that pollen-related food allergens are only nonsensitizing, easily degradable and thus 'irrelevant secondary' allergens should be reconsidered, in particular for pollen-allergic patients with chronic atopic dermatitis.

## Acknowledgments

The author wishes to thank Christof Ebner, Beatrice Jahn-Schmid, Astrid Radakovics, Eva Maria Schimek, Renate Steiner and Bettina Zwölfer. This work was supported by the Fonds zur Förderung der wissenschaftlichen Forschung (FWF, SFB-F1807-B04) and Biomed, Austria.

## References

1. Amiot PL, Kemeny DM, Zachary C, Parkes P, Lessof MH. Oral allergy syndrome (OAS): symptoms of IgE-mediated hypersensitivity to foods. *Clin Allergy* 1987;**17**:33–42.
2. Ballmer-Weber BK, Vieths S, Luttkopf D, Heuschmann P, Wuthrich B. Celery allergy confirmed by double-blind, placebo-controlled food challenge: a clinical study in 32 subjects with a history of adverse reactions to celery root. *J Allergy Clin Immunol* 2000;**106**:373–378.
3. Kleine-Tebbe J, Vogel L, Crowell DN, Hausteiner UF, Vieths S. Severe oral allergy syndrome and anaphylactic reactions caused by a Bet v 1-related PR-10 protein in soybean, SAM22. *J Allergy Clin Immunol* 2002;**110**:797–804.
4. Wuthrich B. Birch pollen allergy in relation with hypersensitivity for various kinds of fruit. *Hautarzt* 1990;**41**:583–584.
5. Wuthrich B, Stager J, Johansson SG. Celery allergy associated with birch and mugwort pollinosis. *Allergy* 1990;**45**:566–571.
6. Ebner C, Hirschwehr R, Bauer L, Breiteneder H, Valenta R, Ebner H et al. Identification of allergens in fruits and vegetables: IgE cross-reactivities with the important birch pollen allergens Bet v 1 and Bet v 2 (birch profilin). *J Allergy Clin Immunol* 1995;**95**:962–969.

7. Kazemi-Shirazi L, Pauli G, Purohit A, Spitzauer S, Frjöschl R, Hoffmann-Sommergruber K et al. Quantitative IgE inhibition experiments with purified recombinant allergens indicate pollen-derived allergens as the sensitizing agents responsible for many forms of plant food allergy. *J Allergy Clin Immunol* 2000;**105**:116–125.
8. Wensing M, Akkerdaas JH, van Leeuwen WA, Stapel SO, Bruijnzeel-Koomen CA, Aalberse RC et al. IgE to Bet v 1 and profilin: cross-reactivity patterns and clinical relevance. *J Allergy Clin Immunol* 2002;**110**:435–442.
9. Karamloo F, Schmitz N, Scheurer S, Foetisch K, Hoffmann A, Hausteiner D et al. Molecular cloning and characterization of a birch pollen minor allergen, Bet v 5, belonging to a family of isoflavone reductase-related proteins. *J Allergy Clin Immunol* 1999;**104**:991–999.
10. Karamloo F, Wangorsch A, Kasahara H, Davin LB, Hausteiner D, Lewis NG et al. Phenylcoumaran benzylic ether and isoflavonoid reductases are a new class of cross-reactive allergens in birch pollen, fruits and vegetables. *Eur J Biochem* 2001;**268**:5310–5320.
11. Breiteneder H, Radauer C. A classification of plant food allergens. *J Allergy Clin Immunol* 2004;**113**:821–830; quiz 831.
12. Vanek-Krebitz M, Hoffmann-Sommergruber K, Laimer Da Camara Machado M, Susani M, Ebner C, Kraft D et al. Cloning and sequencing of Mal d 1, the major allergen from apple (*Malus domestica*), and its immunological relationship to Bet v 1, the major birch pollen allergen. *Biochem Biophys Res Commun* 1995;**214**:538–551.
13. Karamloo F, Scheurer S, Wangorsch A, May S, Hausteiner D, Vieths S. Pyr c 1, the major allergen from pear (*Pyrus communis*), is a new member of the Bet v 1 allergen family. *J Chromatogr B Biomed Sci Appl* 2001;**756**:281–293.
14. Scheurer S, Pastorello EA, Wangorsch A, Kastner M, Hausteiner D, Vieths S. Recombinant allergens Pru av 1 and Pru av 4 and a newly identified lipid transfer protein in the in vitro diagnosis of cherry allergy. *J Allergy Clin Immunol* 2001;**107**:724–731.
15. Breiteneder H, Hoffmann-Sommergruber K, O'Riordan G, Susani M, Ahorn H, Ebner C et al. Molecular characterization of Api g 1, the major allergen of celery (*Apium graveolens*), and its immunological and structural relationships to a group of 17-kDa tree pollen allergens. *Eur J Biochem* 1995;**233**:484–489.
16. Hoffmann-Sommergruber K, O'Riordan G, Ahorn H, Ebner C, Laimer Da Camara Machado M, Pühringer H et al. Molecular characterization of Dau c 1, the Bet v 1 homologous protein from carrot and its cross-reactivity with Bet v 1 and Api g 1. *Clin Exp Allergy* 1999;**29**:840–847.
17. Mittag D, Vieths S, Vogel L, Wagner-Loew D, Starke A, Hunziker P et al. Birch pollen-related food allergy to legumes: identification and characterization of the Bet v 1 homologue in mungbean (*Vigna radiata*), Vig r 1. *Clin Exp Allergy* 2005;**35**:1049–1055.
18. Mittag D, Akkerdaas J, Ballmer-Weber BK, Vogel L, Wensing M, Becker WM et al. Ara h 8, a Bet v 1-homologous allergen from peanut, is a major allergen in patients with combined birch pollen and peanut allergy. *J Allergy Clin Immunol* 2004;**114**:1410–1417.
19. Mittag D, Vieths S, Vogel L, Becker WM, Rihs HP, Helbling A et al. Soybean allergy in patients allergic to birch pollen: clinical investigation and molecular characterization of allergens. *J Allergy Clin Immunol* 2004;**113**:148–154.
20. Hoffmann-Sommergruber K, Demoly P, Cramer R, Breiteneder H, Ebner C, Laimer Da Camara Machado M et al. IgE reactivity to Api g 1, a major celery allergen, in a Central European population is based on primary sensitization by Bet v 1. *J Allergy Clin Immunol* 1999;**104**:478–484.
21. Ballmer-Weber BK, Hoffmann A, Wuthrich B, Luttkopf D, Pompei C, Wangorsch A et al. Influence of food processing on the allergenicity of celery: DBPCFC with celery spice and cooked celery in patients with celery allergy. *Allergy* 2002;**57**:228–235.
22. Neudecker P, Lehmann K, Nerkamp J, Haase T, Wangorsch A, Fotisch K et al. Mutational epitope analysis of Pru av 1 and Api g 1, the major allergens of cherry (*Prunus avium*) and celery (*Apium graveolens*): correlating IgE reactivity with three-dimensional structure. *Biochem J* 2003;**376**:97–107.
23. Asero R. Relevance of pollen-specific IgE levels to the development of Apia-aceae hypersensitivity in patients with birch pollen allergy. *Allergy* 1997;**52**:560–564.
24. De Amici M, Moratti R, Quaglini S, Vignini MA, Mosca M. IgE to Bet v 1/Bet v 2 in UAS. *Allergy* 2002;**57**:181–182.
25. Ballmer-Weber BK, Wuthrich B, Wangorsch A, Fotisch K, Altmann F, Vieths S. Arrot allergy: double-blinded, placebo-controlled food challenge and identification of allergens. *J Allergy Clin Immunol* 2001;**108**:301–307.
26. Ballmer-Weber BK, Wangorsch A, Bohle B, Kaul S, Kundig T, Fotisch K et al. Component-resolved in vitro diagnosis in carrot allergy: does the use of recombinant carrot allergens improve the reliability of the diagnostic procedure? *Clin Exp Allergy* 2005;**35**:970–978.
27. Stevens WJ, Ebo DG, Hagendorens MM, Bridts CH, De Clerck LS. Is the prevalence of specific IgE to classical inhalant aeroallergens among patients with respiratory allergy changing? Evidence from two surveys 15 years apart. *Acta Clin Belg* 2003;**58**:178–182.
28. Schenk S, Breiteneder H, Susani M, Najafian N, Laffer S, Duchene M et al. T cell epitopes of Phl p 1, major pollen allergen of timothy grass (*Phleum pratense*). Crossreactivity with group I allergens of different grasses. *Adv Exp Med Biol* 1996;**409**:141–146.
29. Bohle B, Wagner B, Vollmann U, Buck D, Niggemann B, Szepefalusi Z et al. Characterization of T cell responses to Hev b 3, an allergen associated with latex allergy in spina bifida patients. *J Immunol* 2000;**164**:4393–4398.
30. Jahn-Schmid B, Kelemen P, Himly M, Bohle B, Fischer G, Ferreira F et al. The T cell response to Art v 1, the major mugwort pollen allergen, is dominated by one epitope. *J Immunol* 2002;**169**:6005–6011.
31. Bohle B, Zwolfer B, Fischer GF, Seppala U, Kinaciyan T, Bolwig C et al. Characterization of the human T cell response to antigen 5 from *Vespa vulgaris* (Ves v 5). *Clin Exp Allergy* 2005;**35**:367–373.
32. Fritsch R, Bohle B, Vollmann U, Wiedermann U, Jahn-Schmid B, Krebitz M et al. Bet v 1, the major birch pollen allergen, and Mal d 1, the major apple allergen, cross-react at the level of allergen-specific T helper cells. *J Allergy Clin Immunol* 1998;**102**:679–686.
33. Bohle B, Radakovics A, Jahn-Schmid B, Hoffmann-Sommergruber K, Fischer GF, Ebner C. Bet v 1, the major birch pollen allergen, initiates sensitization to Api g 1, the major allergen in celery: evidence at the T cell level. *Eur J Immunol* 2003;**33**:3303–3310.



34. Bohle B, Radakovics A, Luttkopf D, Jahn-Schmid B, Vieths S, Ebner C. Characterization of the T cell response to the major hazelnut allergen, Cor a 1.04: evidence for a relevant T cell epitope not cross-reactive with homologous pollen allergens. *Clin Exp Allergy* 2005;**35**:1392–1399.
35. Jahn-Schmid B, Radakovics A, Luttkopf D, Scheurer S, Vieths S, Ebner C et al. Bet v 1142–156 is the dominant T-cell epitope of the major birch pollen allergen and important for cross-reactivity with Bet v 1-related food allergens. *J Allergy Clin Immunol* 2005;**116**:213–249.
36. Bohle B, Zwolfer B, Heratizadeh A, Jahn-Schmid B, Antonia YD, Alter M et al. Cooking birch pollen-related food: divergent consequences for IgE- and T cell-mediated reactivity in vitro and in vivo. *J Allergy Clin Immunol* 2006;**118**:242–249.
37. Jensen-Jarolim E, Wiedermann U, Ganglberger E, Zurcher A, Stadler BM, Boltz-Nitulescu G et al. Allergen mimotopes in food enhance type I allergic reactions in mice. *FASEB J* 1999;**13**:1586–1592.
38. Vieths S, Reindl J, Müller U, Hoffmann A, Hausteiner D. Digestibility of peanut and hazelnut allergens investigated by a simple in vitro procedure. *Eur Food Res Technol* 1999;**209**:379–388.
39. Wiche R, Gubesch M, König H, Fotisch K, Hoffmann A, Wangorsch A et al. Molecular basis of pollen-related food allergy: identification of a second cross-reactive IgE epitope on Pru av 1, the major cherry (*Prunus avium*) allergen. *Biochem J* 2005;**385**:319–327.
40. Schimek EM, Zwolfer B, Briza P, Jahn-Schmid B, Vogel L, Vieths S et al. Gastrointestinal digestion of Bet v 1-homologous food allergens destroys their mediator-releasing, but not T cell-activating, capacity. *J Allergy Clin Immunol* 2005;**116**:1327–1333.
41. Reekers R, Busche M, Wittmann M, Kapp A, Werfel T. Birch pollen-related foods trigger atopic dermatitis in patients with specific cutaneous T-cell responses to birch pollen antigens. *J Allergy Clin Immunol* 1999;**104**:466–472.
42. Haselden BM, Kay AB, Larche M. Immunoglobulin E-independent major histocompatibility complex-restricted T cell peptide epitope-induced late asthmatic reactions. *J Exp Med* 1999;**189**:1885–1894.
43. Jankiewicz A, Aulepp H, Baltes W, Bogl KW, Dehne LI, Zuberbier T et al. Allergic sensitization to native and heated celery root in pollen-sensitive patients investigated by skin test and IgE binding. *Int Arch Allergy Immunol* 1996;**111**:268–278.
44. Schocker F, Luttkopf D, Müller U, Thomas P, Vieths S, Becker WM. IgE binding to unique hazelnut allergens: identification of non pollen-related and heat-stable hazelnut allergens eliciting severe allergic reactions. *Eur J Nutr* 2000;**39**:172–180.
45. Scheurer S, Lauer I, Foetisch K, Moncin MS, Retzek M, Hartz C et al. Strong allergenicity of Pru av 3, the lipid transfer protein from cherry, is related to high stability against thermal processing and digestion. *J Allergy Clin Immunol* 2004;**114**:900–907.
46. Bohle B, Hirt W, Nachbargauer P, Ebner H, Ebner C. Allergy to millet: another risk for atopic bird keepers. *Allergy* 2003;**58**:325–328.
47. Asero R. Effects of birch pollen-specific immunotherapy on apple allergy in birch pollen-hypersensitive patients. *Clin Exp Allergy* 1998;**28**:1368–1373.
48. Asero R. How long does the effect of birch pollen injection SIT on apple allergy last? *Allergy* 2003;**58**:435–438.
49. Herrmann D, Henzgen M, Frank E, Rudeschko O, Jäger L. Effect of hypo-sensitization for tree pollinosis on associated apple allergy. *J Invest Allergol Clin Immunol* 1995;**5**:259–267.
50. Hansen KS, Khinchi MS, Skov PS, Bindslev-Jensen C, Poulsen LK, Malling HJ. Food allergy to apple and specific immunotherapy with birch pollen. *Mol Nutr Food Res* 2004;**48**:441–448.
51. Bolhaar ST, Tiemessen MM, Zuidmeer L, van Leeuwen A, Hoffmann-Sommergruber K, Bruijnzeel-Koomen CA et al. Efficacy of birch-pollen immunotherapy on cross-reactive food allergy confirmed by skin tests and double-blind food challenges. *Clin Exp Allergy* 2004;**34**:761–769.
52. Bousquet J. Sublingual immunotherapy: from proven prevention to putative rapid relief of allergic symptoms. *Allergy* 2005;**60**:1–3.
53. Wilson DR, Lima MT, Durham SR. Sublingual immunotherapy for allergic rhinitis: systematic review and meta-analysis. *Allergy* 2005;**60**:4–12.
54. Marogna M, Spadolini I, Massolo A, Canonica GW, Passalacqua G. Clinical, functional, and immunologic effects of sublingual immunotherapy in birch pollinosis: a 3-year randomized controlled study. *J Allergy Clin Immunol* 2005;**115**:1184–1188.
55. Till SJ, Francis JN, Nouri-Aria K, Durham SR. Mechanisms of immunotherapy. *J Allergy Clin Immunol* 2004;**113**:1025–1034; quiz 1035.
56. Ma Y, Gadermaier G, Bohle B, Bolhaar S, Knulst A, Markovic-Housley Z et al. Mutational analysis of amino acid positions crucial for IgE-binding epitopes of the major apple (*Malus domestica*) allergen, Mal d 1. *Int Arch Allergy Immunol* 2006;**139**:53–62.
57. Crimi E, Voltolini S, Gianiorio P, Orengo G, Troise C, Brusasco V et al. Effect of seasonal exposure to pollen on specific bronchial sensitivity in allergic patients. *J Allergy Clin Immunol* 1990;**85**:1014–1019.
58. Karlsson MG, Davidsson A, Hellquist HB. Increase in CD4<sup>+</sup> and CD45RO<sup>+</sup> memory T cells in the nasal mucosa of allergic patients. *Apmis* 1994;**102**:753–758.
59. Wosinska-Becler K, Plewako H, Hakansson L, Rak S. Cytokine production in peripheral blood cells during and outside the pollen season in birch-allergic patients and non-allergic controls. *Clin Exp Allergy* 2004;**34**:123–130.
60. Bohle B, Schwihla H, Hu HZ, Friedl-Hajek R, Sowka S, Ferreira F et al. Long-lived Th2 clones specific for seasonal and perennial allergens can be detected in blood and skin by their TCR-hypervariable regions. *J Immunol* 1998;**160**:2022–2027.
61. Seddon B, Tomlinson P, Zamoyska R. Interleukin 7 and T cell receptor signals regulate homeostasis of CD4 memory cells. *Nat Immunol* 2003;**4**:680–686.
62. Bagnasco M, Mariani G, Passalacqua G, Motta C, Bartolomei M, Falagiani P et al. Absorption and distribution kinetics of the major *Parietaria judaica* allergen (Par j 1) administered by non-injectable routes in healthy human beings. *J Allergy Clin Immunol* 1997;**100**:122–129.
63. Ohashi Y, Nakai Y, Tanaka A, Kakinoki Y, Ohno Y, Sakamoto H et al. Seasonal rise in interleukin-4 during pollen season is related to seasonal rise in specific IgE for pollens but not for mites. *Acta Otolaryngol* 1998;**118**:243–247.